Solution: Price Index by Dorbish and Bowley's Method

p_0	q_0	p_1	q_1	$p_0 q_0$	$p_0 q_1$	$p_1 q_0$	$p_1 q_1$
80	3	100	4	240	320	300	400
35	2	45	3	70	105	90	135
25	2	30	3	50	75	60	90
50	30	54	35	1500	1750	1620	1890
35	2	40	3	70	105	80	120
				1930	2355	2150	2635

$$P_{01}^{DB} = \frac{\frac{\sum p_1 q_0}{\sum p_0 q_0} + \frac{\sum p_1 q_1}{\sum p_0 q_1}}{2} \times 100$$

$$= \frac{1}{2}\left[\frac{2150}{1930} + \frac{2635}{2355}\right] \times 100$$

$$= \frac{1}{2}[1.1139 + 1.1188] \times 100$$

$$= \frac{1}{2}[2.2327] \times 100$$

$$= 1.1164 \text{ x } 100 = 111.64$$

Example 6.9

Compute Marshall – Edgeworth price index number for the following data by taking 2016 as base year.

Items sold out in a men's wear	2016		2017	
	Prices	Quantity	Prices	Quantity
Shirts	700	150	900	175
Pants	1000	100	1200	150
Sandals	500	70	600	100
Shoes'	1500	50	1800	60
Belts	400	100	600	150
Watches	1200	300	1500	250

Solution: Price Index by Marshall-Edgeworth Method

p_0	q_0	p_1	q_1	$p_0 q_0$	$p_0 q_1$	$p_1 q_0$	$p_1 q_1$
700	150	900	175	105000	122500	135000	157500
1000	100	1200	150	100000	150000	120000	180000
500	70	600	100	35000	50000	42000	60000
1500	50	1800	60	75000	90000	90000	108000
400	100	600	150	40000	60000	60000	90000
1200	300	1500	250	360000	300000	450000	375000
				715000	772500	897000	970500

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